forest/non-forest land and improve the quality of forest cover on another 5 million ha. Funding for the scheme to come from the Plan outlay and convergence with the Mahatma Gandhi National Rural Employment Guarantee Scheme, the compensatory afforestation management and planning authority, and the national afforestation programme.

The Union government to provide 90 per cent of the funds for implementing the scheme in the north-eastern states, and 75 per cent of the funds for other states. State governments to meet the balance requirement.

Forest policy in India is victim of the fact that decision-making on environmental issues in India is quite fragmented. While the ministry of environment and forests implements various central laws– including the Forest Conservation Act 1980– the state governments, too, have powers in this domain. There are other ministries and decision-making authorities that are involved as well. If that is not all, the Supreme Court and other (Green) tribunals, too, are active participants in the ongoing story of India's forests. Other stakeholders– tour operators, villages located in forest areas and various illegal, interest groups such as timber smugglers and poachers also manage to exercise influence through the political system.

All this makes for serious collective action problems. As a result, formulating coherent policies for forest and wildlife conservation and implementing them effectively are well-nigh impossible.

4.7 MINERAL RESOURCES

The mineral resources of India encompass a wide range of products that are necessary for a modern developed economy. There are, according to the Geological Survey of India, 50 important minerals and 400 major sites where these minerals occur. (Out of total land area of 3.28 million sq.kms, hard rock area covers 2.42 million sq.kms.). These can be divided into four categories as follows:

1) Minerals of which India's exportable surplus can dominate the world market; to this category belong iron-ore and mica;
2) Minerals of which the exportable surplus forms an important factor; these include manganese ore, bauxite, gypsum and others;
3) Minerals in which it appears that the country is self-sufficient, like coal, sodium salts, glass sand, phosphates, bauxite, etc.;
4) Minerals for which India has to depend largely or entirely on foreign markets like copper, nickel, petroleum, lead, zinc, tin, mercury, platinum, graphite, etc.

The various minerals can also be classified into three categories based on their nature and end use. These three categories are: (i) Fuels like coal, lignite, natural gas and petroleum; (ii) Metallic minerals like bauxite, iron-ore, manganese, etc.; (iii) Non-metallic minerals like phosphorite, graphite, gypsum, limestone, mica, etc.

4.7.1 Features of Minerals

Minerals provide a base for the rapid industrialisation of the economy. The changeover to an open market economy has opened further avenues for faster industrial growth and greater requirement of minerals, "besides the fact that the geological setting of the country holds great promise for a boom in mineral productions." [Expert opinion is that given the size of deposits in South Africa and Australia, large reserves can be expected in India also (because of similar geological structures). A recent Price Waterhouse Report has identified India as the most promising mining location worldwide.]

Notwithstanding this, India's spending on exploration is only about 0.8 per cent of the global spending with private sector contributing only 3 per cent of that. There are a few essential aspects that need to be worked into a proper mineral policy.

1) The mineral resources are very unevenly distributed.

   - The Great Plains of Northern India are almost entirely devoid of any known deposits of economic minerals.
   - Jharkhand and Odisha on the north-eastern parts of peninsular India possess large concentration of mineral deposits, accounting for nearly three-fourths of the country's coal deposits and containing highly rich deposits of iron-ore, manganese, mica, bauxite and radioactive minerals.
   - Mineral deposits are also scattered over the rest of the peninsular India and in parts of Assam and Rajasthan.
   - With the implementation of the new UN lawon the Exclusive Economic Zone (EEZ), new areas in deep seas, hitherto unexplored, would become available to us. These areas are believed to contain rich deposits of oil, gas, manganese, nickel, cobalt and copper.

2) The country is deficient in certain minerals like crude oil or petroleum;

   Nearly 80 per cent of the present domestic demand is being met by imports. In view of the rising prices of these minerals in international markets, it would be necessary, on the one hand, to curb their growing use in the economy, and, on the other hand, sustained efforts should be made to augment the domestic sources of supply of these minerals.

3) There are minerals which are lucrative foreign exchange earners. Efforts should be made to devise a suitable policy to have a proper utilisation of these minerals keeping in view the national interests.

4) Due to paucity of funds, the mining industry is mired in obsolete technology, for decades. India's mining sector is in trouble. The sector's troubles stem from profit-seeking, poor regulation, and galloping demand from countries such as China. This has led to courts stepping in and banning all mining activity in a region. Environmental clearance, and the human cost involved, are other matters the industry has just begun to deal with.

4.7.2 New Mineral Policy, 2008

A committee was set up in the late-2005 under the chairmanship of Anwar-ul-Hoda. The New Mineral Policy, 2008 has incorporated most of the recommendations made in the Hoda Committee Report.

i) The new policy manifestly recognises that private sector will be the main source of investment in reconnaissance and exploration. It takes a pragmatic view of the risks associated with the business of finding/producing minerals and therefore ensures security of tenure and rights of transferability of various awards– reconnaissance permits (RPs), prospecting licenses (PLs) and mining leases (MLs).

ii) The new policy provides for introducing Long Area Prospecting Licences to help the investors achieve economies of scale. These licences will be given only with regard to non-bulk minerals which, from the investors' point of view, is a high-risk, high reward area.

iii) The policy emphasise that mining is a standalone industrial activity, it can thrive in conjunction with value-addition activities.

iv) Export policies will be formulated after taking stock of the mineral inventories and short, medium and long-term needs of the country. It is stated in the policy document that efforts shall be made to export minerals in value-added forms as far as possible.

v) The policy envisages steps to be taken by the governments to facilitate financing of mine development and exploration which are integral to the mining projects.

vi) There is also a plan to develop appropriate capital market structures to attract risk investment in survey and prospecting. Flow of 'risk funds'– from capital market and venture funds– will be eased with policy interventions. The public private partnership model would be useful for building mining infrastructure.

vii) The policy seeks to bring certain level of uniformity in mineral administration in the country.

Further, acting on the recommendations of the Ashok Chawla committee, the Government sought to seek outside help to increase the pace of mapping of potential mineral reserves in India to meet a government target that aimed to complete prospecting for natural resources in the five years beginning April 1, 2012.

India's mining sector is set for a cleanup. Some of the steps in that regard are:

- Setting up of coal regulator to set methodologies for price fixation by Coal India;
- Competitive bidding for allocation of captive blocks to private companies;
- Improved coal supply for power plants as coal and power ministries iron out differences over issues in new supply pacts;
- Higher production by Coal India as fresh clearances start coming after Prime Minister Office's (PMO's) intervention;